

Composition and the Algebra of Mechanisms

One object type, a small operator set, and convex duality underneath

Peter Cotton · Working draft v0.1 · 2026

Abstract

Market mechanisms — scoring rules, cost-function market makers, constant-function AMMs, parimutuels, opinion pools — are usually presented as a list. The more useful structure is how they compose. This note sets out an algebra: every stage consumes and emits one object type (a distributional belief), threads wealth as its state, and is generated by a convex function, with Legendre duality as the central involution. Under these conventions Savage’s characterisation of proper scoring rules, Hanson’s sequentialisation into a market maker, and the cost-function/CFMM duality are three readings of one potential, and a small operator set — sequentialise, pool, conjugate, ensemble, residual — closes over the catalogue. A worked pipeline chains an elicitation market into a calibration critic through the probability integral transform, with wealth as the proper-scoring gradient, and converges numerically. Properness under composition is settled by the injectivity theorems of the transformation literature; conservation across stages and residual (boosting) markets remain open.

1. One object type: a distributional belief

The operational discipline is borrowed from the [skaters](#) time-series interface ([Cotton 2026](#)), a dependency-free successor to [timemachines](#) ([Cotton 2021](#)). A skater has the contract

```
dists, state = f(y, state)           # dists : list[Dist] for horizons 1..k
```

where every payload is a `Dist`, a weighted Gaussian mixture exposing `.mean`, `.std`, `.quantile`, `.logpdf`, `.cdf`. Transforms, ensembles, and search all consume and emit that same type, and that closure is what lets a handful of operators suffice.

Every mechanism in the catalogue already speaks the same language:

Mechanism	As a map on distributions
scoring rule	$(\text{Dist}, \text{outcome}) \rightarrow \text{reward}$
LMSR / cost-function maker	$\text{state} \rightarrow \text{Dist}$ (prices are implied probabilities)
parimutuel	$\text{stakes} \rightarrow \text{Dist}$ (pool fractions are implied probabilities)
opinion pool	$[\text{Dist}] \rightarrow \text{Dist}$
PIT / calibration	$(\text{Dist } F, x) \rightarrow z = \Phi^{-1}(F(x))$

So the mechanisms are already a category whose objects are distributional beliefs. What the skater discipline supplies is the operational signature that turns them into composable pipeline stages, with wealth as the threaded state:

```

payouts, aggregate, wealth = M(reports, wealth, outcome)
# reports      : list[Dist]    participants' beliefs      (skater's y)
# wealth       : state         incentive-bearing          (skater's state)
# aggregate    : Dist          price / implied prob       (skater's dists)

```

A market maker is a skater whose state is the inventory vector $\mathbf{q}$ and whose emitted `Dist` is the price. Hanson's construction is exactly the skater move: sequentialising a proper scoring rule over a wealth state turns it into a market maker (Theorem 2).

2. The operators

Skaters expose a small operator set; each lifts to a relationship between mechanisms.

Conjugation. `conjugate(skater, transform)` runs a model in a transformed coordinate and maps back (difference, standardize, ema; all invertible). It appears twice in the mechanism world, and the coincidence of names is not accidental:

- *Statistical.* “One market determines a transformation, another tests whether it is uniform” is literally conjugation. Market A elicits a predictive CDF F ; that F is the transform; the z-stream $z = \Phi^{-1}(F(x))$ is the conjugated coordinate; Market B trades against the null that z is iid $N(0, 1)$.
- *Convex-analytic.* The cost-function/CFMM duality is the Legendre conjugate of the potential (Proposition 3): the trading-function market is the cost-function market viewed in dual coordinates.

Ensemble. The precision-weighted skater ensemble ($\propto 1/\text{MSE}$) is an inverse-variance linear opinion pool; the Bayesian ensemble (log-likelihood with a complexity penalty) is a logarithmic opinion pool (Genest and Zidek 1986). The nearest-the-pin pool of the companion paper is a wealth-weighted ensemble whose mixing weights are themselves a market, updated by realised score.

Residual chaining. One stage models what the previous stage got wrong: a correction (boosting) market on the residual stream. This operator is named here and analysed as an open direction in the companion research note on conservation and boosting markets.

Sequentialise. The Hanson functor, `score` $\rightarrow$ market maker, with wealth as state (Theorem 2).

Pool. The batch functor, `score` $\rightarrow$ parimutuel: elicitation in one shot rather than sequentially.

Spec / grammar. Skaters serialise a pipeline to data and search over it. The mechanism analogue is a market over pipelines: the architecture itself becomes the thing being elicited.

3. The generator: convex functions under Legendre duality

Underneath the operators is a single generating object: convex functions, with Legendre duality as the central involution.

- A convex G generates a strictly proper scoring rule; the induced divergence is G 's Bregman divergence (Savage 1971; Banerjee, Merugu, et al. 2005).
- A convex potential C generates a no-arbitrage cost-function market maker with prices ∇C (Abernethy et al. 2013).

- The conjugate C^* generates the trading function of a CFMM (Frangillo et al. 2024).

Operations on convex functions then become operations on mechanisms:

Operation on the potential	Effect on the mechanism
Legendre conjugation $C \mapsto C^*$	cost-function maker $\mapsto$ CFMM (the dual reading)
scaling $C \mapsto bC$	the liquidity knob b (depth vs subsidy)
inf-convolution $C_1 \sqcap C_2$	merging two liquidity sources into one book (Rockafellar 1970; Moreau 1965; Bhaskara et al. 2023)
a state monad over wealth	static elicitation $\mapsto$ dynamic, competitive market

The inf-convolution row is also the risk-sharing literature’s composition law: the market aggregate of several agents’ convex risk measures is their infimal convolution, with the Pareto allocation as minimiser and the common subgradient as the clearing price (Barrieu and El Karoui 2005; Jouini et al. 2008).

This is the sense in which there is an algebra: objects are distributions; the generators are convex potentials; the morphisms are sequentialise, pool, conjugate, ensemble, and residual; and composition closes because every stage emits the same Dist the next stage consumes.

4. The classic results, stated

The claim “convex functions generate the mechanisms” is a chain of theorems. Conventions here are reward-form (higher is better); the repository’s implementations use the loss form, i.e. the negation. Outcomes lie in $\{1, \dots, n\}$, a report is p in the simplex Δ , and the expected score of reporting p when one believes q is $S(p; q) = \sum_i q_i S(p, i)$. A scoring rule is *proper* if truthful reporting is optimal, $S(q; q) \geq S(p; q)$ for all p, q , and *strictly proper* if equality forces $p = q$.

Theorem 1 (characterisation of proper scoring rules; Savage (1971), McCarthy (1956)). *A regular scoring rule S is proper iff there is a convex function $G : \Delta \rightarrow \mathbb{R}$ with*

$$S(p, i) = G(p) + \langle G'(p), e_i - p \rangle, \quad G'(p) \in \partial G(p),$$

where e_i is the i -th unit vector. It is strictly proper iff G is strictly convex, and then $G(p) = S(p; p)$ is the expected score of a truthful forecaster (the generalised entropy) and $G'(p)$ is its gradient.

Proof sketch. ($\Leftarrow$) Because $\sum_i q_i (e_i - p) = q - p$, the expected score collapses to an affine function of the belief, $S(p; q) = G(p) + \langle G'(p), q - p \rangle$. Hence

$$S(q; q) - S(p; q) = G(q) - G(p) - \langle G'(p), q - p \rangle = D_G(q, p) \geq 0,$$

the Bregman divergence of G , non-negative by the supporting-hyperplane inequality and zero only at $q = p$ when G is strictly convex. ($\Rightarrow$) Define $G(p) := S(p; p)$. Properness says $G(q) =$

$\max_p S(p; q)$, a pointwise maximum of functions affine in q ; a pointwise sup of affine functions is convex, so G is convex. The maximiser at $q = p$ exhibits $S(p, \cdot)$ as a supporting hyperplane of G at p , which is the displayed formula with $G'(p) \in \partial G(p)$. Uniqueness of the maximiser is equivalent to strict convexity. ■

The content of Theorem 1 is the dictionary *proper scoring rule* $\leftrightarrow$ *convex function* $\leftrightarrow$ *Bregman divergence* (Gneiting and Raftery 2007; Banerjee, Guo, et al. 2005). The three classics are three choices of G :

Score	Generator $G(p) = S(p; p)$	Bregman divergence $D_G(q, p)$
logarithmic	$\sum_i p_i \log p_i$ (negative entropy)	$\text{KL}(q \ p)$
Brier (quadratic)	$\ p\ _2^2$	$\ q - p\ _2^2$
spherical	$\ p\ _2$	angular separation of q, p

Theorem 2 (scoring rule $\Rightarrow$ market maker; Hanson (2007), Abernethy et al. (2013)). Let $R = G$, the convex generator of Theorem 1 read as a regulariser, and define the cost function as its Legendre–Fenchel conjugate

$$C(\mathbf{q}) = \sup_{p \in \Delta} (\langle p, \mathbf{q} \rangle - R(p)), \quad \text{prices } \nabla C(\mathbf{q}) = \arg \max_p (\cdot) \in \Delta.$$

Then a trader moving inventory $\mathbf{q} \rightarrow \mathbf{q}'$ pays $C(\mathbf{q}') - C(\mathbf{q})$, prices are a valid probability vector, the market is arbitrage-free because C is convex, and the market maker’s worst-case loss is the range of R ,

$$\text{worst-case loss} = \sup_p R(p) - \inf_p R(p).$$

Taking $R(p) = b \sum_i p_i \log p_i$, b times the log-score generator, gives $C(\mathbf{q}) = b \log \sum_i e^{q_i/b}$, Hanson’s LMSR, with worst-case loss $b \log n$.

Proof sketch. Conjugate duality gives $\nabla C(\mathbf{q}) = \arg \max_p$, which lies in Δ , and $\langle \nabla C, \mathbf{1} \rangle = 1$ by the simplex constraint, so prices are probabilities. Telescoping costs over any trade sequence returning to $\mathbf{q}$ sum to zero, ruling out arbitrage. The maker’s loss after the market settles on outcome i is bounded by how far R can move between its extremes, $\sup R - \inf R$; for b times negentropy on Δ that range is $b \log n$. The conjugacy $C = R^*$ is the formal content of “the log score, run sequentially, is LMSR.” ■

Proposition 3 (cost-function maker $\leftrightarrow$ CFMM is Legendre duality; Frongillo et al. (2024)). A constant-function market maker holds reserves and accepts trades preserving a concave trading function φ ; a cost-function maker prices via convex C . The two are the same object read through $C \leftrightarrow C^*$: the reachable reserve set of the CFMM is the (sub)gradient image of C , and the price the CFMM quotes equals ∇C at the corresponding inventory.

Proposition 4 (the probability integral transform; Rosenblatt (1952), Dawid (1984)). If X has continuous CDF F then $U = F(X) \sim \text{Uniform}(0, 1)$, and hence $z = \Phi^{-1}(U) \sim N(0, 1)$.

Proof. $\Pr(F(X) \leq u) = \Pr(X \leq F^{-1}(u)) = F(F^{-1}(u)) = u$. ■

Proposition 4 is the formal warrant for §5: a correctly elicited predictive distribution makes the z-stream standard normal, so any detectable departure is exactly a miscalibration a critic can be paid to find. (For discrete forecasts one uses the randomised or mid-PIT; the demo uses the latter.)

5. Worked composition: an elicitation market feeding a calibration critic

The cleanest non-trivial pipeline chains two mechanisms through a feedback loop (runnable: [examples/sim_pipeline.py](#)):

1. *Elicitation market (the transformation)*. A pool of forecasters each report a predictive distribution; the wealth-weighted linear opinion pool is a single predictive distribution F_t . F_t is the transformation, the coordinate change that should turn outcomes into noise.
2. *Calibration critic (test for uniformity)*. Apply the probability integral transform $u_t = F_t(x_t)$; if F_t is right the u_t are Uniform(0,1) and $z_t = \Phi^{-1}(u_t)$ is standard normal. The critic measures how far the z-stream departs from uniform; its detectable edge is exactly the aggregate’s miscalibration.
3. *Feedback*. Each forecaster’s wealth is updated by its log score, the proper-scoring gradient, so wealth flows to the calibrated reports, F_t sharpens toward the truth, and the critic’s edge collapses.

This is a market-native adversarial pair: a generator (the transform market) against a critic (the uniformity market), with a proper score as the loss and wealth as the gradient. It is the z-stream mechanism of the microprediction vision, and z-streams ran on a live forecasting platform ([Cotton 2022](#)).

The demo makes the loop quantitative. Five forecasters (two calibrated, plus a bull, a bear, and an overconfident report) against an iid $N(0, 1)$ stream:

- z-stream uniformity error (total-variation distance from uniform): equal-weighted aggregate 0.165, converged aggregate 0.068;
- bias (mean z) stays near zero throughout — the symmetric bull and bear cancel, so the mean never reveals the problem. It is the shape of the PIT that is wrong and then fixed, which is why the critic tests uniformity of the whole z-stream rather than a moment or two;
- wealth on the two calibrated reports climbs $0.41 \rightarrow 1.00$ as the loop runs.

6. Composition as engineering, and what remains open

Read this way, market and scoring mechanisms are composable pipeline stages with a property ordinary pipeline stages lack: each stage carries its own incentive, so the pipeline is self-correcting. One market picks a transformation; a second market is paid to find structure the first one left behind; wealth, not a hand-tuned loss, routes credit between them. A forecasting stack, a feature pipeline, or a data-quality monitor can each be posed as a chain of elicitation and test markets glued by the PIT.

Three questions attach to the operator set.

1. *Properness under composition*. Settled in the transformation literature: properness is preserved under any transformation of forecast and outcome, and strict properness iff the

- transform is injective (Allen et al. 2023, Prop. 4; Pic et al. 2025, Prop. 1). The PIT is the injective case; a learned, non-invertible transform loses strictness. The Markov-kernel (stochastic-channel) extension is Theorem 2 of the companion point-cloud paper.
2. *Incentive compatibility of the chain.* Each market is individually truthful; the composite can admit strategic play across stages (cf. the equilibrium analysis of the pot split in §2 of the companion paper).
 3. *Conservation and residual markets.* Wealth is the threaded state; a well-formed pipeline should conserve the elicited *edge*, not just wealth, across stage seams, and a residual (boosting) chain of markets remains to be constructed. Both are set out with calibrated novelty verdicts in the companion research note on conservation and boosting.

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- Runnable demo: `examples/sim_pipeline.py`. Companion papers and notes: *Scoring Point-Cloud Distributional Submissions; conservation and boosting; the relationship map*.